

Panda: A Pretrained Forecast Model for Chaotic Dynamics

Reading Report

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Contents

1	Problem Statement	3
2	Background Theory	3
2.1	Dynamical Systems and Phase Space	3
2.2	Attractors and Strange Attractors	4
2.3	Chaos and Lyapunov Exponents	4
2.4	Fractal Geometry and Correlation Dimension	5
2.5	Ergodic Theory and Invariant Measures	5
2.6	Takens' Embedding Theorem	6
2.7	Koopman Operator and Extended Dynamic Mode Decomposition	6
2.8	Synchronization and Skew-Product Systems	6
3	Dataset Construction	7
3.1	Motivation	7
3.2	Founding Population	7
3.3	Evolutionary Search	7
3.4	Dataset Diversity	8
3.5	Augmentations	8
3.6	Held-Out Test Set	8
4	Model Architecture	8
4.1	Design Philosophy	8
4.2	Patching	9
4.3	Dynamics Embedding	9
4.4	Temporal Attention	9
4.5	Channel Attention	10
4.6	Prediction Head and Training	10
5	Results	11

5.1	Zero-Shot Forecasting of Held-Out Chaotic Systems	11
5.2	Ablations	11
5.3	Zero-Shot Forecasting of Experimental Data	12
5.4	The Dynamical Systems Scaling Law	12
5.5	Emergent PDE Forecasting Capability	13
5.6	Interpretable Internal Representations	13
5.7	Long-Term Behavior of Predictions	14
6	Limitations and Future Directions	14

1 Problem Statement

Chaotic dynamical systems appear throughout science and engineering. Some of the fluid flows, neuronal circuits, electronic oscillators, and mechanical systems exhibit property of sensitive dependence on initial conditions. Any small error in the initial state grows exponentially over time, making long-term point forecasting fundamentally limited. This is not a limitation of any particular model, rather it is a mathematical consequence of the phenomenon of chaos.

Prior data-driven approaches to forecasting chaos fall into two categories. **Specialized local models** are trained on a single dynamical system and forecast future states of that same system. They perform well in domain but are useless for any other system. **General time series foundation models** such as Chronos or TimesFM are trained on large databases of real-world time series (retail, finance, weather) and applied zero-shot, but they treat chaotic trajectories the same way they treat stock prices, with no physics information of dynamical systems theory. Both approaches fail to answer this: is there something universal about chaotic dynamics that a model could learn across many different systems and then transfer to unseen ones?

This is the problem Panda addresses. The goal is a global forecast model for chaotic dynamics; one trained on a large and diverse collection of chaotic systems that generalizes zero-shot to systems it has never encountered. This is an instance of out-of-domain generalization, which is fundamentally harder for chaos than for domains like image classification because two chaotic systems may have geometrically incomparable attractors with no obvious shared surface features.

The paper makes four concrete contributions: (1) a dataset of 2×10^4 novel chaotic ordinary differential equations (ODEs) generated via evolutionary search, (2) a pretrained multivariate transformer (Panda) that achieves competitive zero-shot forecasts on both synthetic held-out systems and real experimental data, (3) evidence that three architecture choices motivated by dynamical systems theory each contribute independently, and (4) an emergent capability to forecast partial differential equations (PDEs) despite being trained exclusively on low-dimensional ODEs.

2 Background Theory

This section contains theoretical concepts required to understand the dataset construction, architecture design, and evaluation methodology of Panda.

2.1 Dynamical Systems and Phase Space

A *continuous-time dynamical system* is specified by an ODE (where the function does not depend on time)

$$\dot{x} = f(x), \quad x \in \mathbb{R}^n, \quad (1)$$

where $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a smooth vector field. The solution starting at x_0 is written $\phi^t(x_0)$ and called the *flow*. The space $\mathbb{R}^n$ in which the state x lives is the *phase space* or *state space*. A trajectory is the curve $\{\phi^t(x_0) : t \geq 0\}$ traced by a single initial condition.

Fixed points are states x^* satisfying $f(x^*) = 0$. The trajectory starting at a fixed point stays there forever. Local stability of a fixed point is determined by the eigenvalues of

the Jacobian $Df(x^*)$: eigenvalues with negative real part correspond to stable directions, eigenvalues with positive real part to unstable directions. A fixed point is *asymptotically stable* (an attractor) if all eigenvalues have negative real part.

Limit cycles are isolated closed trajectories, i.e. periodic orbits that are the long-time limit of nearby trajectories. They require at least two dimensions. A stable limit cycle is an attractor for nearby initial conditions.

The Poincaré–Bendixson theorem states that in a two-dimensional continuous dynamical system, any bounded trajectory must converge to a fixed point or a closed orbit. A direct consequence is that *chaos is impossible in two-dimensional autonomous (where f is time-independent) continuous systems*. At least three dimensions are required.

2.2 Attractors and Strange Attractors

The ω -limit set of a point x_0 is

$$\omega(x_0) = \bigcap_{T \geq 0} \overline{\{\phi^t(x_0) : t \geq T\}}, \quad (2)$$

the set of all accumulation points of the forward orbit. An *attractor* is a compact invariant set A such that $\omega(x_0) \subseteq A$ for all x_0 in an open neighbourhood of A , and no strictly smaller closed set has this property.

A *strange attractor* is an attractor that is (a) bounded, (b) aperiodic (no trajectory repeats), and (c) fractal (non-integer dimension). The canonical example is the Lorenz attractor, arising from

$$\begin{aligned} \dot{x} &= \sigma(y - x), \\ \dot{y} &= x(\rho - z) - y, \\ \dot{z} &= xy - \beta z, \end{aligned} \quad (3)$$

with $\sigma = 10$, $\rho = 28$, $\beta = 8/3$. At these parameters, trajectories spiral around one lobe of the butterfly for a while, then switch to the other lobe, then switch back, never repeating and never escaping to infinity.

2.3 Chaos and Lyapunov Exponents

The defining property of chaos is *sensitive dependence on initial conditions*: two trajectories starting at distance $|\delta(0)|$ apart separate as

$$|\delta(t)| \approx |\delta(0)|e^{\lambda_1 t}, \quad (4)$$

where $\lambda_1 > 0$ is the *maximum Lyapunov exponent*. The system is chaotic if and only if $\lambda_1 > 0$.

An n -dimensional system has a full *Lyapunov spectrum* $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$. For a dissipative chaotic attractor: $\lambda_1 > 0$ (divergence), $\lambda_2 = 0$ (motion along the flow), and the remaining exponents are negative (contraction onto the attractor). The sum satisfies

$$\sum_{i=1}^n \lambda_i = \langle \nabla \cdot f \rangle_\mu, \quad (5)$$

which is negative for dissipative systems.

The *Lyapunov time* $T_{\text{Lyap}} = 1/\lambda_1$ is the natural predictability horizon, the timescale over which errors double. Beyond a few Lyapunov times, any finite-precision point forecast becomes meaningless.

The *Kaplan–Yorke conjecture* estimates the fractal dimension of the attractor from the Lyapunov spectrum:

$$D_{KY} = j + \frac{\sum_{i=1}^j \lambda_i}{|\lambda_{j+1}|}, \quad (6)$$

where j is the largest integer such that $\sum_{i=1}^j \lambda_i \geq 0$. For the Lorenz attractor $D_{KY} \approx 2.06$.

In practice, λ_1 is estimated from data using the *Rosenstein estimator*: find nearest-neighbour pairs of trajectory points and track how their separation grows over time. The slope of $\langle \log d(t) \rangle$ versus t gives λ_1 .

2.4 Fractal Geometry and Correlation Dimension

The *Grassberger–Procaccia correlation dimension* D_2 is a computable proxy for the fractal dimension of an attractor. Given a trajectory $\{x_1, \dots, x_T\}$, define the correlation integral

$$C(r) = \frac{2}{T(T-1)} \sum_{i < j} \mathbf{1}[\|x_i - x_j\|_2 < r]. \quad (7)$$

For small r , $C(r) \sim r^{D_2}$, so $D_2 = d \log C(r) / d \log r$, estimated by the slope on a log-log plot. For the Lorenz attractor $D_2 \approx 2.06$.

2.5 Ergodic Theory and Invariant Measures

A *measure-preserving flow* preserves a probability measure μ : $\mu(\phi^t(A)) = \mu(A)$ for all measurable sets A . For a chaotic attractor, the physically relevant measure is the *SRB* (*Sinai–Ruelle–Bowen*) *measure*, which is the long-run histogram of a typical trajectory:

$$\mu_{\text{SRB}} = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \delta_{\phi^t(x)} dt. \quad (8)$$

An ergodic system satisfies the *Birkhoff ergodic theorem*: time averages equal space averages μ -almost surely,

$$\lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T g(\phi^t(x)) dt = \int g d\mu. \quad (9)$$

This means a single long trajectory samples the entire invariant measure.

Pesin’s formula connects information theory to dynamics:

$$h_{KS} = \sum_{\lambda_i > 0} \lambda_i, \quad (10)$$

where h_{KS} is the Kolmogorov–Sinai metric entropy — the rate of information production by the system. Each new trajectory from the same system contributes new samples from the same measure at rate h_{KS} , but once the measure is well-estimated, additional

trajectories provide diminishing new information. New systems, by contrast, provide qualitatively new topological information (winding structure, attractor geometry) that no number of trajectories from an existing system can supply. This is the theoretical foundation for the scaling law discovered in Section 5.4.

2.6 Takens' Embedding Theorem

A fundamental obstacle in real experiments is that one can observe only a scalar function of the full state. Takens' theorem says this is sufficient.

Theorem 1 (Takens 1981). *Let M be a compact smooth d -dimensional manifold, $\phi : M \rightarrow M$ a smooth diffeomorphism, and $y : M \rightarrow \mathbb{R}$ a smooth observation function. The delay embedding map*

$$\Phi_{y,\tau,k}(x) = (y(x), y(\phi^\tau(x)), \dots, y(\phi^{(k-1)\tau}(x))) \quad (11)$$

is an embedding (injective with injective derivative) for $k \geq 2d + 1$, for generic pairs (ϕ, y) and generic delay τ .

The consequence is that the delay-embedded trajectory in $\mathbb{R}^k$ is diffeomorphic to the original trajectory on M : same attractor topology, same Lyapunov exponents, same invariant measure. This justifies two key architecture choices in Panda: *patching* (a patch of P consecutive observations is a delay vector) and the *time-delay augmentation* used in dataset construction.

2.7 Koopman Operator and Extended Dynamic Mode Decomposition

The Koopman operator provides a way to linearize nonlinear dynamics at the cost of moving to an infinite-dimensional function space. For a measure-preserving flow with measure μ , the Koopman operator $\mathcal{K}^t$ acts on observable functions $g \in L^2(M, \mu)$ by

$$(\mathcal{K}^t g)(x) = g(\phi^t(x)). \quad (12)$$

Although ϕ^t is nonlinear, $\mathcal{K}^t$ is always *linear*. Koopman eigenfunctions φ_k satisfy $\mathcal{K}^t \varphi_k = e^{\lambda_k t} \varphi_k$ and evolve as pure exponentials along trajectories.

In practice, the Koopman operator is approximated in a finite dictionary $\Psi(x) = [\psi_1(x), \dots, \psi_N(x)]^\top$ via *extended dynamic mode decomposition* (eDMD): find the best linear map $\tilde{K}$ such that $\Psi(x_{n+1}) \approx \tilde{K} \Psi(x_n)$. Standard dictionary choices are polynomial features (Taylor-motivated) and random Fourier features (kernel approximation). As $N \rightarrow \infty$, eDMD converges to the true Koopman operator. The dynamics embedding in Panda is precisely an eDMD dictionary applied patch-wise.

2.8 Synchronization and Skew-Product Systems

A *skew-product system* couples a driver f_a to a response f_b asymmetrically:

$$\begin{aligned} \dot{x} &= f_a(x, t), \\ \dot{y} &= \kappa_b f_b(y, t) + \kappa_a f_a(x, t). \end{aligned} \quad (13)$$

The driver evolves independently. By the theory of chaotic synchronization (Pecora and Carroll, 1990), the response either synchronizes to the chaotic driver ($y(t) \rightarrow x(t)$) or

exhibits its own chaos — both outcomes preserve a positive Lyapunov exponent in the combined system. The asymmetry is essential: symmetric mutual coupling can stabilize both subsystems into a periodic orbit, destroying chaos. The skew-product coupling guarantees that chaoticity is preserved under recombination.

3 Dataset Construction

3.1 Motivation

Real-world chaotic datasets are small, at most a few hundred well-studied systems. Training a global model requires thousands of qualitatively distinct attractors. The authors generate synthetic data. Every generated system is verified to be chaotic, stationary, and genuinely distinct from its parents.

3.2 Founding Population

The starting point is a hand-curated dataset of 129 previously published low-dimensional chaotic ODEs (Gilpin, 2021; 2023), ranging from three to ten dimensions. Each system has been hand-tuned to its chaotic regime and integrated on a standardized timescale based on its largest Lyapunov exponent.

3.3 Evolutionary Search

New systems are generated through three operations.

Mutation. For each system f_a with parameters θ_a , sample perturbed parameters $\theta'_a \sim \mathcal{N}(\theta_a, \sigma)$. This produces systems topologically similar to their parent but with slightly different attractor geometry.

Recombination. Sample two mutated systems f_a (driver) and f_b (response) and form the skew-product system (equations above). The scale factors $\kappa = 1/\sqrt{\mathbb{E}\|f(x, t)\|^2}$ normalize each flow’s RMS magnitude so neither overwhelms the other numerically. Recombination creates genuinely new attractor topologies — hybrid systems that cannot be reduced to either parent.

Selection. Each candidate system is integrated using an implicit fifth-order Runge–Kutta solver (Radau) with tight tolerances ($\text{rtol} = 10^{-9}$, $\text{atol} = 10^{-10}$) and passed through a battery of tests:

- Divergence and fixed-point filtering (transient rejection).
- Chaos 0-1 test: score $K \in [0, 1]$; $K \approx 1$ indicates chaos, $K \approx 0$ indicates quasiperiodicity.
- Near-recurrence test: rejects limit cycles.
- Power spectrum test: rejects trajectories with only a few sharp spectral peaks (quasiperiodic signature).
- Rosenstein estimator: requires $\lambda_1 > 0$.
- KPSS and augmented Dickey–Fuller stationarity tests.

Only systems passing all tests are retained. The result is 2×10^4 verified chaotic, stationary ODEs.

An implicit integrator (Radau) is used because many of the generated systems are stiff — they have components evolving on vastly different timescales. Explicit methods such as RK4 require a stepsize bounded by the fastest timescale, making integration of slow components prohibitively expensive. Radau is A-stable (stability region covers the entire left half-plane) and handles stiffness efficiently.

3.4 Dataset Diversity

The correlation dimension (Grassberger–Procaccia) is approximately preserved between founder and offspring systems (2.09 ± 0.27 vs 2.11 ± 0.23), confirming the absence of a founder effect or mode collapse. The UMAP projection of all 2×10^4 systems shows the parent systems broadly dispersed across the offspring cloud with no clustering, indicating genuine diversity.

3.5 Augmentations

Three augmentations expand the training distribution while preserving dynamical validity, applied to 20% of training trajectories:

- **Time-delay embedding:** $x_i(t) \rightarrow x_i(t - \tau_i)$, $\tau_i \sim \mathcal{U}(1, d_{\text{embed}})$. By Takens’ theorem, the resulting trajectory is diffeomorphic to the original.
- **Convex combinations:** $X \leftarrow MX$ where rows of M are sampled from a Dirichlet distribution. Mixes observation channels while preserving attractor topology.
- **Affine transforms:** $X \leftarrow AX + b$ with random A , b . Rescales and translates the attractor. All dynamical invariants (Lyapunov exponents, correlation dimension) are preserved under diffeomorphisms.

These augmentations are principled in the sense that they create equivalent dynamics in different coordinate systems, not fictitious dynamics.

3.6 Held-Out Test Set

Twenty of the 129 founding systems are withheld entirely from training. The test set is formed by evolving these 20 systems through the same evolutionary procedure, yielding 9,300 held-out skew-product systems. No test system shares genetic lineage with any training system, preventing any form of data leakage through shared ancestry.

4 Model Architecture

4.1 Design Philosophy

Standard time series foundation models with causal decoders tend to “parrot”. They repeat motifs from the context window. For regular time series this is a useful inductive bias. For chaotic trajectories (which never repeat) it is a fundamental failure mode. Panda instead uses an **encoder-only, fixed-horizon** architecture: the full 512-step context is processed simultaneously in a non-causal manner, then a fixed 128-step forecast is produced. This maximises short-term pointwise accuracy — what the authors call “predicting the weather” as opposed to “predicting the climate” (learning the invariant measure).

4.2 Patching

A multivariate trajectory $T \in \mathbb{R}^{C \times T_{\text{len}}}$ is tokenised into non-overlapping patches of length $P = 16$:

$$T_P \in \mathbb{R}^{C \times (T_{\text{len}}/P) \times P}. \quad (14)$$

A 512-step trajectory becomes 32 patches per channel. Patching serves two purposes. Practically, it reduces sequence length by a factor of P , yielding a $P^2 = 256\times$ reduction in attention cost and a $\sim 60\times$ inference speedup over token-wise models. Theoretically, each patch is a delay embedding vector of length P ; by Takens’ theorem, it encodes topological information about the current attractor state. A single timestep carries almost no attractor information; a patch of 16 consecutive observations carries the local dynamical history needed to localise the trajectory on the attractor.

4.3 Dynamics Embedding

Each patch $P \in \mathbb{R}^{C \times 16}$ is lifted to embedding dimension $d_{\text{model}} = 512$ by concatenating three components:

Raw patch (dimension 16): the 16 raw observations.

Random polynomial features $\Phi(P)$ (dimension 120): degree-2 and degree-3 monomials of patch entries, selected from a fixed random index set.

Random Fourier features $F(P)$ (dimension 256): sample $W \in \mathbb{R}^{16 \times 128}$, $b \in \mathbb{R}^{128}$ from a Gaussian and compute

$$F(P) = [\sin(PW + b), \cos(PW + b)] \in \mathbb{R}^{C \times 256}. \quad (15)$$

The full embedding is $E(P) = [P, \Phi(P), F(P)]$.

The weights W and b are *fixed at initialisation and never trained*. This follows the reservoir computing paradigm: random projections into a sufficiently high-dimensional space approximately preserve inner products (Johnson–Lindenstrauss), and training them would risk overfitting the embedding to the specific training attractors. The polynomial and Fourier features together form an eDMD dictionary: polynomial features approximate the Koopman operator for polynomial vector fields, and Fourier features approximate a shift-invariant kernel (Rahimi and Recht, 2007). The transformer then learns an approximate Koopman propagator in this lifted space.

4.4 Temporal Attention

The channel dimension is treated as a batch dimension and self-attention is applied over the $T_{\text{len}}/P = 32$ patches per channel independently. This is univariate temporal mixing. Partial rotary positional encoding (p-RoPE, Barbero et al. 2025) is used with $p = 0.75$, meaning 75% of attention head dimensions carry positional information and 25% are position-agnostic. The wavelength is set to 500.

Positional encoding is essential for temporal attention because the ordering of patches in time carries dynamical meaning. The trajectory at patch 5 is causally related to the trajectory at patch 4 in a way that cannot be ignored.

4.5 Channel Attention

After temporal attention the token sequence is transposed and self-attention is applied over the C channels at each patch position:

$$\text{ChannelAttention}(T_P) = \text{SelfAttention}(T_P^\top), \quad T_P^\top \in \mathbb{R}^{(T_{\text{len}}/P) \times C \times d_{\text{model}}}. \quad (16)$$

No positional encoding is used in channel attention. This makes the operation *permutation equivariant* with respect to channels. Relabelling the state variables does not change the output. This is the correct inductive bias because the coupling between channels is encoded in their values, not their labels. Permutation equivariance also allows the model to generalise from the 3-channel training setup to arbitrary channel counts at test time: the model never learned “there are exactly 3 channels,” it learned “attend between channels based on content similarity.”

Chaotic dynamical systems exhibit strong channel coupling. The variables x, y, z of the Lorenz system are governed by coupled ODEs, and ignoring these couplings causes the forecast to produce trajectories that look correct channel-by-channel but fail to respect the joint attractor geometry. This is demonstrated empirically in Section 5.3: as the coupling strength between channels increases, Panda’s advantage over univariate baselines grows.

Implicit spatiotemporal coupling. Although temporal and channel attention are applied sequentially, their composition implements implicit spatiotemporal coupling. The cross-channel mixing map

$$M_i^{k\ell} = (\mathbf{p}_{TA}^{(k)})^\top \begin{bmatrix} \langle \mathbf{p}_1^{(k)}, \tilde{A}_{CA} \mathbf{p}_1^{(\ell)} \rangle & \cdots \\ \vdots & \ddots \end{bmatrix} \mathbf{p}_{TA}^{(\ell)} \quad (17)$$

shows that patch j of channel ℓ influences patch i of channel k through a bilinear operation mixing both temporal and channel information.

4.6 Prediction Head and Training

Each attention block applies temporal attention, then channel attention, then a feedforward network, each with RMSNorm and residual connections. After 8 such blocks, processed tokens are averaged over the patch dimension and mapped linearly to a length-128 forecast for all channels.

During training, 3 channels are randomly sampled from each trajectory to enable efficient batching across systems of different dimensionality. At inference time the full channel set is used; channel equivariance ensures no mismatch.

Two training modes are used:

Forecasting mode: predict the next 128 steps given 512 steps of context. Loss is MSE.

Masked language model (MLM) mode: randomly mask 50% of patches and reconstruct them. This forces the model to use global context — understanding what patches are consistent with each other — rather than purely local extrapolation. MLM pretraining improves short-horizon accuracy but reduces autoregressive rollout performance (see Section 5.2). A separate MLM checkpoint is therefore maintained for the completions task.

All models are trained with AdamW, learning rate 10^{-3} , cosine schedule with 10% warmup, gradient clipping at 1.0.

5 Results

5.1 Zero-Shot Forecasting of Held-Out Chaotic Systems

Setup. Panda (21M parameters) is evaluated zero-shot on 9,300 held-out systems across a range of prediction horizons. Six context windows are used per system. Metrics are reported as median and semi-interquartile range.

Baselines. Five baselines are evaluated: Chronos 20M (causal univariate decoder, trained on real-world time series), Chronos 20M SFT (Chronos fine-tuned on Panda’s training data — the fairest comparison, isolating architecture from data), TimesFM 200M (patch-based decoder, univariate), Time MOE 50M (sparse mixture-of-experts, univariate), and DynaMix (multivariate recurrent model, trained on the 129 founding systems).

Short-horizon results. At prediction horizon 128, Panda achieves median sMAPE 27.6 versus 30.3 for Chronos SFT and 36.0 for Chronos 200M (Table 8 in the paper). The gap to Chronos SFT is entirely architectural — same training data, but Panda’s encoder-only non-causal processing and channel attention provide decisive advantage. Panda also outperforms TimesFM 200M ($10\times$ larger) and Time MOE 50M ($2.4\times$ larger), demonstrating efficiency.

Horizon generalisation. Panda was trained with a prediction horizon of 128 steps. At test time, window autoregression (rolling its own predictions back as context) extends forecasting to 512 steps — $4\times$ the training horizon — with graceful degradation. This is possible because predicted trajectories, even when diverging from the true trajectory pointwise, remain on the attractor. Errors are bounded by attractor geometry rather than growing without bound.

Statistical significance. Wilcoxon signed-rank tests with Holm-Sidák correction confirm all improvements are statistically significant ($p < 10^{-47}$ against all baselines on MSE at $L = 128$).

5.2 Ablations

Three architectural components are ablated:

Channel attention is the largest single contributor. Removing it (univariate variants) consistently increases error, with the gap growing with prediction horizon. Coupling effects accumulate over time, so the benefit of explicitly modeling them compounds.

MLM pretraining improves short-horizon accuracy significantly. The model benefits from seeing past and future patches simultaneously during pretraining, developing a global understanding of trajectory structure. However, MLM hurts autoregressive rollout: the global attention patterns learned during MLM are suboptimal for local one-step-ahead extrapolation.

Dynamics embedding (polynomial features and random Fourier features) helps without MLM and specifically improves long-horizon rollout via PolyEmbed (polynomial features only). With MLM it slightly reduces short-horizon performance but improves long rollout.

5.3 Zero-Shot Forecasting of Experimental Data

Panda is tested zero-shot on three real experimental datasets: a double pendulum (positions and momenta, 4 channels), *C. elegans* eigenworms (leading independent components of body posture, 3–5 channels), and a network of 28 randomly coupled electrical oscillators (voltage recordings).

These datasets exhibit nonstationarity, measurement noise, and missing values not present in training. Despite this, Panda outperforms Chronos SFT on all three systems. The transfer works for three reasons. First, Takens’ theorem is coordinate-free: voltage recordings, body posture projections, and pendulum tip positions are all valid observation coordinates, and the theorem guarantees the delay embedding reconstructs the attractor regardless of the physical meaning of the channels. Second, the random Fourier and polynomial features are smooth functions of the input, making the embedding noise-robust. Third, the training distribution is diverse enough that the real experimental attractors fall within the range of dynamical behaviors the model has encountered.

The electronic circuit experiment provides the most theoretically interesting result. The log ratio $\log(\text{sMAPE}_{\text{Panda}}/\text{sMAPE}_{\text{Chronos-SFT}})$ is plotted as a function of coupling strength and prediction horizon. At low coupling strength the two models are comparable. As coupling strength increases, Panda’s advantage grows monotonically. At strong coupling and long horizons Panda is dramatically better. This confirms that channel attention captures dynamical coupling, not merely statistical correlation.

5.4 The Dynamical Systems Scaling Law

Experimental design. Eight training datasets are constructed with identical total timepoints but varying allocation: one extreme has $N_{\text{sys}} \approx 20,000$ systems with one trajectory each; the other has $N_{\text{sys}} \approx 156$ systems with 128 trajectories each. A separate model is trained on each dataset and evaluated on the same 9,300 held-out systems.

Result. Performance scales as a power law in the number of unique systems:

$$\mathbb{E}[\text{sMAPE}] \approx 28.12 + 151.67 \cdot N_{\text{sys}}^{-0.4195}, \quad R^2 = 0.9896. \quad (18)$$

More unique systems consistently outperforms more trajectories per system, even with identical total data volume.

Theoretical grounding. This is a direct empirical consequence of ergodic theory. By Birkhoff’s theorem, a single sufficiently long trajectory from an ergodic system already samples the full invariant measure. Additional trajectories provide more samples from the same measure — diminishing returns, governed by Pesin’s formula. New systems provide qualitatively different topological information: distinct attractor geometry, winding structure, and fractal dimension. This information cannot be obtained from any number of trajectories of existing systems. The scaling law says diversity in attractor topology is the correct axis along which to scale data for a global chaotic dynamics model.

This scaling law is distinct from standard neural scaling laws (Kaplan et al. 2020), which scale total data volume at fixed composition. Here total data volume is held fixed and composition is varied. The steeper exponent (-0.42 versus -0.07 to -0.10 for language models) indicates faster improvement per additional system early on, but also faster saturation. Performance has not yet saturated at 2×10^4 systems.

5.5 Emergent PDE Forecasting Capability

Panda is applied zero-shot to two weakly turbulent PDEs: the Kuramoto–Sivashinsky (KS) equation (1D, 64 Fourier modes) and the von Kármán vortex street (VKVS, vorticity field reduced to 512 PCA components via a 256×128 spatial grid simulation). Neither PDE was seen during training.

Despite this, Panda outperforms two in-distribution baselines (FNO and DeepONet trained on the specific PDE) across all prediction horizons. Qualitatively, Panda correctly predicts nonlinear events: flame front merging and creation in KS, and vortex pinch-off in VKVS. These baselines predict smooth dynamics and miss discrete topological events.

Three mechanisms explain the transfer. First, dissipative PDEs possess *inertial manifolds*, which are finite-dimensional invariant manifolds in function space on which the long-time dynamics reduce to a finite-dimensional ODE. For KS on a domain of length 100, the inertial manifold dimension is approximately 10–20, comparable to what Panda was trained on. Second, Ruelle and Takens (1971) showed that spatiotemporal chaos arises generically through bifurcations from low-dimensional dynamics, so PDE attractors carry the geometric fingerprints of low-dimensional chaos. Third, channel attention captures spatial coupling between PDE grid points exactly as it captures dynamical coupling between ODE state variables — the model has no knowledge that channels represent space rather than state.

The operator learning baselines (FNO, DeepONet) are limited to one-step prediction rolled out autoregressively, accumulating error as $e^{\lambda_1 t}$. Panda predicts 128 steps simultaneously, directly optimising multi-step accuracy, which makes it more robust to rollout error accumulation.

5.6 Interpretable Internal Representations

To probe what Panda has learned, the authors feed two-tone sinusoids at frequencies f_1 , f_2 independently swept over $[2\pi, 5\pi]$ and measure the average row-wise entropy of the *attention rollout* $\tilde{A}$ (the product of all temporal attention matrices across layers).

The resulting plot shows complex multiscale structure at diagonal bands and curved ridges corresponding to nonlinear frequency combinations $f_1 + f_2$, $f_1 - f_2$, $2f_1$, and so forth. This is *nonlinear resonance* — the same phenomenon exhibited by forced pendulums and kicked rotors. A linear system would respond only at f_1 and f_2 ; the nonlinear combinations are evidence that Panda has internalised something about nonlinear frequency mixing intrinsic to chaotic dynamics.

A univariate ablation (same architecture without channel attention) shows no such structure — the frequency response is flat and featureless. Channel attention is the mechanism enabling nonlinear resonance, because frequency mixing happens between coupled channels, not within a single channel’s time series.

Direct visualisation of temporal attention maps reveals four qualitative structures: Toeplitz (shift-invariant, for stationary ergodic systems), block (for systems with distinct dynamical phases or intermittency), selector (sharp concentration on a few key patches near turning points or bifurcation-like events), and hybrid combinations. Some maps resemble classical recurrence plots from nonlinear dynamics, suggesting the model implicitly implements recurrence analysis.

5.7 Long-Term Behavior of Predictions

At sufficiently long horizons, all models fail at pointwise prediction. The relevant question becomes how they fail, and what their long-run distributional properties look like.

Panda mean regresses. The model’s predictions collapse toward zero (the instance-normalized mean) at very long horizons. This is confirmed quantitatively in Table 17: when metrics are computed only on the tail of a $L_{\text{pred}} = 3584$ forecast (the last 2048 of 3584 steps), Panda’s KL divergence is 15.25 ± 2.46 , the worst of all models. The transformer context window is fixed at 512 steps; beyond it, the model has no new information and hedges to the safe prediction.

Chronos parrots. Chronos repeats context motifs at long horizons. For ergodic chaotic systems with frequent near-recurrences (unstable periodic orbits), this accidentally produces good distributional statistics.

DynaMix sustains. The recurrent architecture maintains a hidden state that integrates information from the entire trajectory history with no context window limit, giving the best long-horizon distributional fidelity ($\text{KL} = 3.50$ at the tail). The tradeoff is clear: transformers win at short-horizon pointwise accuracy, recurrent models win at long-horizon distributional fidelity. They are complementary architectures.

Despite mean regression, Panda maintains the correct maximum Lyapunov exponent distribution out to $25\times$ its training horizon, demonstrating that its forecasts remain chaotic with the right divergence rate long after pointwise accuracy has failed.

6 Limitations and Future Directions

The paper identifies two principal limitations, which are potential open research directions.

Low-dimensional training. Panda is trained exclusively on 3 to 10 dimensional ODEs. Generalisation to high-dimensional systems (large PDEs, fully turbulent flows) is limited because the inertial manifold argument breaks down at high Reynolds numbers, and channel attention does not exploit known spatial structure. The proposed fix is to allow channel attention to receive *custom attention masks* reflecting known coupling sparsity (spatial locality in PDEs, block structure in network systems). This would make the spatial inductive bias explicit rather than relying on emergent learning.

Pretraining objective. MLM pretraining improves short-horizon accuracy but degrades long-horizon rollout. The optimal pretraining objective for dynamical systems is an open question. Candidates include next-patch prediction (emphasises local extrapolation), masked completion (emphasises global attractor structure), or predicting invariant quantities directly (Lyapunov exponents, correlation dimension). The authors call this “a basic question that necessitates further progress in SciML.”

Long-horizon stability. Mean regression at long horizons is a fundamental architectural weakness of fixed-context transformers. Hybrid transformer-recurrent architectures, or pretraining objectives that directly optimise long-horizon distributional fidelity, are natural directions. From a numerical analysis perspective, the long-horizon stability problem has structural analogies to the stability of numerical ODE integrators: a model that mean-regresses is analogous to a dissipative numerical scheme that damps high-frequency

modes. Developing regularisation strategies informed by integrator stability theory is an open direction.

Spatiotemporal extension. The scaling law implies that expanding training to spatiotemporal systems (diverse PDEs with different spatial geometries, boundary conditions, and spectral structures) would yield large gains. The evolutionary dataset generation framework is extensible to spatiotemporal systems.